This is a range of clinical disorder resulting in severe cases to marasmus or kwashiorkor.

c) Fats and essential fatty acids

Fat intake should be 35 % E depending on the physical activity of the child from age 6 months to 2 years. Linoleic acid is the most important essential fatty acid for an infant. DHA levels in red blood cells and neural tissues help in improving visual acuity and cognitive performance of infants. Both cow's milk and mother's milk satisfy the requirements of EFA. EFA requirement of young children is 3% E which can be satisfied by 19g per day visible fat.

d) Minerals - Calcium and phosphorus

Rapid growth requires 500 mg of calcium and 750 mg of phosphorus with a ratio 1:15. Adequate prenatal nutrition supply a store of bone minerals to prevent rickets provide postnatal care furnishers a liberal supply of calcium and phosphorus. Large percentage of calcium from breast milk is retained by the infant. When sufficient calcium is not supplied to the infant, there motor development is delayed. Ca: P ratio of 1.2: 1 as in cow's milk is lower compared to 2: 1 in human milk.

i) Iron

RDA of iron for an infant is 46µg/kg bodyweight starting from 3 months. At birth, body contains 80mg/ kg. This is about 3 times that of an adult. During the first four months, blood volume doubles and concentration of iron in haemoglobin falls to about half, that present at birth.

ii) Zinc

High levels are present in colostrum and it promotes normal growth. Zinc is necessary for normal brain development.

e) Vitamins

i) Vitamin A

The RDA for retinol is 350µg. Daily intake of Vitamin A by Indian infants through breast milk is about 140µg during the first 6 months of life. After six months egg yolk is supplemented in the infant's diet.

ii) Vitamin D

It is essential for utilisation and retention of calcium and phosphorus. A good supply of vitamin D during pregnancy benefits the mother and it helps satisfactory development of the infant. The vitamin D requirement of child is 400 I.U. if there is minimal exposure to sunlight.

Table 2.6 ICMR recommended dietary allowances for infants 2010

Nutrient	Months	
	0-6	6-12
Energy Kcals	92	80
Protein gm	1.16	1.69
Fats gm	-	19
Calcium mg	500	500
Iron mg	46 µg /kg	05
Vitamin A		
Retinol µg or	350	350
Carotene µg	2800	2800
Thiamine mg	0.2	0.3
Riboflavin mg	0.3	0.4
Niacin equivalent µg /kg	710	650
Pyridoxine mg	0.1	0.4
Ascorbic acid mg	25	25
Dietary folate µg	25	25
Vitamin B12 µg	0.2	0.2
Magnesium mg	30	45
Zinc mg	-	-



Activity : 5

Collect birth weight and height of 5 infants.

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